

XTENZIUM DIGITAL AGENCY

Document Title: Research, Analysis, & Solution Specification

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Prepared By: M. Umar Siddiqui, Senior Partner

Research & Analysis Overview

The strategic analysis of JE Academy's operational workflow indicates a critical need for an automated, zero-clutter digital infrastructure. The academy currently faces challenges in maintaining real-time communication with parents, tracking granular academic progress, and managing administrative bottlenecks.

The Solution: The proposed Student Record Management (SRM) system is a dual-purpose architectural framework combining a highly optimized, public-facing informational portal with a secure, enterprise-grade Role-Based Access Control (RBAC) internal operations dashboard. By integrating direct WhatsApp/SMS APIs, enforcing strict token-based video tracking, and consolidating multi-sibling profiles under unified parent accounts, the system will lower administration friction, reduce cognitive overhead for backend modules, and completely eliminate technical onboarding lifecycles.

1. Key Features

- **Public-Facing Informational Portal:** A single-page, globally accessible architecture detailing institutional values, active programmatic offerings, dynamic curriculum guidelines, faculty professional profiles, physical location coordinate integrations, and secure lead-generation inquiry portals.
- **Automated Communication Pipeline:** Instantaneous WhatsApp and SMS triggers linked directly to attendance and grade uploads, ensuring parents are notified of absences and monthly results the exact moment database entries are recorded.
- **Unified Parent Tracking Hub:** A secure, consolidated access point where parents with multiple children enrolled in the academy can monitor all siblings through a single set of credentials.
- **Strict Video Tracking for Late Enrollments:** An enforced engagement matrix for students joining mid-session. The system tracks actual watch-time on video lectures, actively preventing false completion logs caused by merely opening and closing the media.
- **Dynamic Progress Reporting:** Automated generation of comprehensive downloadable performance telemetry, aggregating monthly test reports, half-yearly exams, and final exam marks.

2. Scope of Work

The engineering pipeline spans from baseline structural design to continuous application monitoring. The core development encompasses:

- **Database & Logic Provisioning:** Setting up the core database indexing, RBAC logic provisioning, cross-device layout adaptation, and final application deployment execution.
- **Portal Development:** Assembly of the frontend public UI alongside four segregated administrative environments (Super, Marks, Attendance, Admissions) and two end-user tiers (Student, Parent).
- **API & Tracking Integration:** Hooking in real-time messaging APIs (WhatsApp/SIM network failovers) and engineering the client-side video tracking logic for late-enrolled students.
- **Ongoing Mitigation:** Comprehensive continuous application monitoring, instant security patching updates, real-time bug mitigation response, continuous system backups, and database normalization.

3. Management Roles and Responsibilities

The system relies on micro-targeted interfaces where security rules determine feature visualization, ensuring employees interact solely with components mapping explicitly to their active mandates.

Role Profile	System Permissions & Responsibilities
Super Admin	Features absolute systemic master access rights, including comprehensive administrative monitoring controls, dynamic sub-administrator lifecycle profiling, master raw log audits, and system-wide visibility vectors. Can override and modify any entry within the SRM.
Admissions Admin	Acts as a purpose-built processing hub facilitating incoming student registration pipelines and generating high-fidelity structural PDF telemetry files for institutional output records. Responsible for uploading and deleting student records, which automatically triggers the creation and dispatch of parent IDs and passwords.
Attendance Admin	Operates a hyper-focused module serving real-time class registration management. Displays active daily class roster configurations alongside single-click check-in tools. Actions trigger immediate, automated absence alerts to parents.
Marks Admin	Utilizes a segregated operational pipeline engineered exclusively for scholastic grade entries. Controls bulk mark uploads, explicit student tier evaluations, and historical modification records.

Student (End-User)

Protected via end-to-end token validation profiles. Has access to direct client-side views enabling instantaneous study guide reference, dynamic lecture file access, and course assignment submission modules.

Parent (End-User)	Enforces an absolute read-only operational profile tracking real-time student attendance matrices, cross-referenced marks metrics, and aggregated monthly progress trends. Access is enabled exclusively via secure credential matching keys supplied directly by JE Academy management.
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4. Key Factors

- **Commercial Framework Waiver:** Standard comprehensive system core development fees (0 PKR) are fully waived to eliminate upfront capital strain.
- **Retention Linkage:** The waiver of development charges is legally linked to the strict execution of a minimum 6-month high-intensity maintenance commitment.
- **Maintenance Investment:** Phase 1 (Months 1-6) requires a 20,000 PKR/Month investment, transitioning to Phase 2 (Month 7 Onwards) at 10,000 PKR/Month for preventative security scanning and upkeep.
- **Infrastructure Dependencies:** JE Academy assumes absolute sole responsibility for purchasing, validating, and sustaining the web domain address, as well as securing standalone, high-performance web hosting infrastructure.

5. Key Points

- **Strict Design Freeze:** Upon formal visual or structural layout sign-off by the client on user interface wireframes, an immediate and absolute Design Freeze is enacted. No visual architecture modifications will be fulfilled post-approval without executing a standalone billable change request.
- **Content Provision:** JE Academy must explicitly supply all core textual materials, administrative logos, initial system database rosters, lecture note media, and assignment outlines prior to development completion.
- **Accessibility Metrics:** The public-facing portal is engineered with light execution speed footprints and structured to secure clear visual positioning and layout legibility on legacy mobile equipment.